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ORIGINAL PAPER

# The Role of Interfacial Reactions in Determining Plasma–Liquid Chemistry

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**Abstract** In this work, we investigate the production of highly oxidative species in solutions exposed to a self-pulsed corona discharge in air. We examine how the properties of the target solution (pH, conductivity) and the discharge power affect the discharge stability and the production of $H_2O_2$. Indigo carmine, a common organic dye, is used as an indicator of oxidative strength and in particular, hydroxyl radical ($OH^{\cdot}$) production. The observed rate of indigo oxidation in contact with the discharge far exceeds that predicted from reactions based on concentrations of species measured in the bulk solution. The generation of $H_2O_2$ and the oxidation of indigo carmine indicate a high concentration of highly oxidizing species such as $OH^{\cdot}$ at the plasma–liquid interface. These results indicate that reactions at the air plasma–liquid interface play a dominant role in species oxidation during direct non-equilibrium atmospheric pressure plasma treatment.

**Keywords** Plasma activated water (PAW) · Reactive oxygen species · Corona discharge · Indigo carmine · Non-equilibrium atmospheric pressure plasma (NEAPP)

## Introduction

Devices to produce "cold" (non-equilibrium) atmospheric-pressure plasma (NEAPP) are gaining increasing attention for a wide range of applications. In addition to promising work in the field of plasma medicine [1–4], NEAPP has also been shown to effectively decontaminate food [5] and surfaces [6, 7], degrade organic pollutants [8–10], and open

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new routes for nanomaterials synthesis [11–13]. In many of these applications, plasma–liquid interfaces are present and mediate the transport of reactive species generated by the plasma to their intended targets in the liquid phase. While the effects of plasma-generated reactive oxygen and nitrogen species (RONS) on specific biological targets have been well studied [14–19], less is known about the role of the plasma–liquid interface in mediating these effects.

It is known that NEAPP discharges in air produce chemically active species in an adjacent liquid phase. The rate at which plasma-generated species are taken up by the adjacent liquid is determined by their gas-phase concentrations and aqueous solubilities, generally expressed as Henry's law constants, as previously discussed [20]. The accumulation of relatively stable RONS in solution, specifically hydrogen peroxide ($H_2O_2$), nitrite ($NO_2^-$), and nitrate ($NO_3^-$), has been well established [8, 21–26]. Sometimes the term 'plasma activated water' (PAW) is used to describe these solutions [25, 27, 28].

Evidence for the air plasma-generated formation of highly reactive aqueous phase radicals and neutral species ($OH^\cdot$, $NO^\cdot$, $O_2^{\cdot-}$, $O_3$, ONOOH) and their activity on biological targets has also been demonstrated. Hydroxyl radicals are of particular importance in biomedical applications as they are readily formed through reactions with water and are highly reactive. In plasma systems involving humid gas or direct contact with aqueous solutions, $OH^\cdot$ is formed through electron impact dissociation (1), dissociative excitation transfer to $H_2O$ from $N_2(A)$ (2), photodissociation of water (3), and charge exchange of incident positive ions on the liquid surface (4) [29–31]:

$$e^- + H_2O \rightarrow OH^\cdot + H^\cdot + e^- \tag{1}$$

$$N_2(A) + H_2O \rightarrow OH^\cdot + H^\cdot + N_2(X) \tag{2}$$

$$h\nu + H_2O_{aq} \rightarrow H^\cdot_{aq} + OH^\cdot_{aq} \tag{3}$$

$$2H_2O_{aq} + M^+ \rightarrow M + H_3O^+_{aq} + OH^\cdot_{aq} \tag{4}$$

The first two mechanisms produce $OH^\cdot$ in the gas phase that may partition into the liquid phase; reactions (3) and (4) produce $OH^\cdot$ in solution. Once in the liquid phase, the lifetimes of plasma generated species are determined by their reactivity and the local concentration of reaction partners. In general, $OH^\cdot$ reacts rapidly and unselectively with ionic species and biologically relevant molecules, with high rate constants ranging from $10^7$ to $10^{10}$ $M^{-1}$ $s^{-1}$ [32–35]. For example, the rate constant for the reaction of $OH^\cdot$ with $NO_2^-$ is $1.1 \times 10^{10}$ $M^{-1}$ $s^{-1}$; the rate constant for reaction with $H_2O_2$ is lower at $2.7 \times 10^7$ $M^{-1}$ $s^{-1}$ [36]. Due to their high reactivity, plasma-generated hydroxyl radicals are not expected to diffuse far beyond a thin boundary layer, on the order of $\sim$100 nm to tens of microns, near the plasma–solution interface before encountering a reaction partner [20, 31].

The reaction of two hydroxyl radicals to form $H_2O_2$ is also significant. The three-body mediated combination of two hydroxyl radicals to form $H_2O_2$ in the gas phase through reaction (5) has been reported to be the primary source of $H_2O_2$ in PAW, although $H_2O_2$ may also be formed through the reaction of two aqueous hydroxyl radicals (6) at sufficiently high local concentrations [31, 37, 38]:

$$OH^\cdot + OH^\cdot + M \rightarrow H_2O_2 + M \tag{5}$$

$$OH^{\cdot}_{aq} + OH^{\cdot}_{aq} \rightarrow H_2O_{2aq} \tag{6}$$

where M is a collision partner (e.g. $N_2$, $H_2O$). The rate of $H_2O_2$ generation is thus sometimes used as a relative indicator of hydroxyl radical ($OH^{\cdot}$) generation [39].

Hydroxyl radicals and other highly reactive species may also be produced through reactions between plasma-generated species in the liquid phase. Among the most important pathways for $OH^{\cdot}$ formation via liquid phase reactions is the formation of peroxynitrous acid (O=NOOH, pKa = 6.8) under acidic conditions through reaction (7). Peroxynitrous acid is unstable and decomposes through reactions (8) and (9) (with the indicated branching ratio), resulting in the formation of $OH^{\cdot}$ [21, 40].

$$NO_2^- + H_2O_2 + H^+ \rightarrow O = NOOH + H_2O \tag{7}$$

$$O = NOOH \rightarrow NO_2^{\cdot} + OH^{\cdot} \sim 30\,\% \tag{8}$$

$$O = NOOH \rightarrow HNO_3 \rightarrow NO_3^- + H^+ \sim 70\,\% \tag{9}$$

Thus there are multiple pathways for the generation of $OH^{\cdot}$ in acidic aqueous solutions in contact with NEAPP. Plasma-mediated reactions above the liquid surface can create $OH^{\cdot}$ that reacts at or very near the liquid surface. In addition, the decomposition of O=NOOH or other reactions either near the surface or in the 'bulk' of the liquid can generate $OH^{\cdot}$. In this

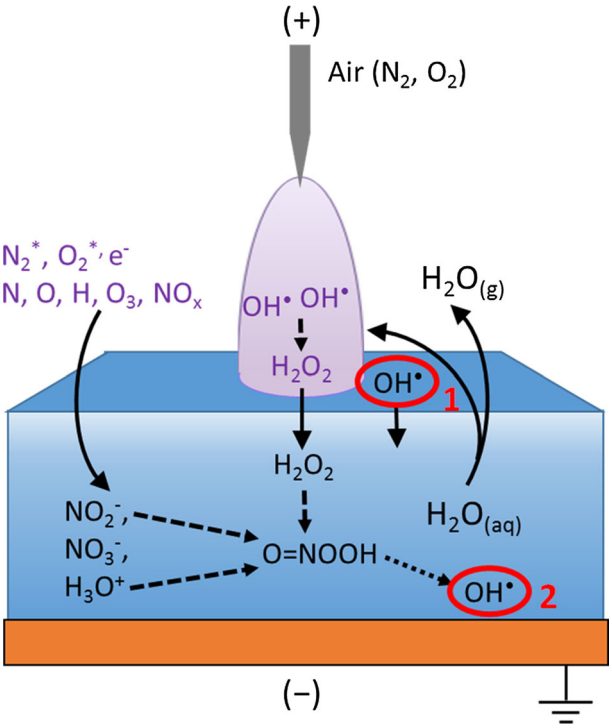
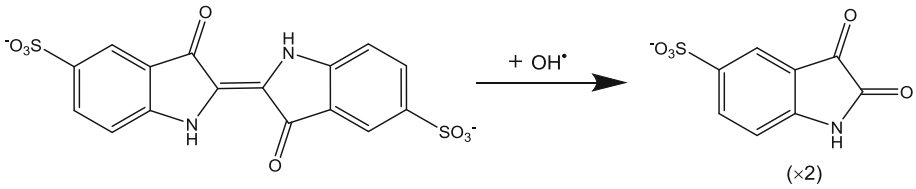



**Fig. 1** Sources and pathways of $OH^{\cdot}$ produced by positive corona discharge above water. Two sources of $OH^{\cdot}$ are emphasized: (*1*) $OH^{\cdot}$ generated from plasma-induced reactions at or near the plasma–solution interface, and (2) the decomposition of O=NOOH in solution under acidic conditions

work, we investigate the relative importance of near-surface and bulk sources in the generation of $OH^{\cdot}$ by a positive corona discharge in air, as illustrated in Fig. 1.

The concentration of $OH^{\cdot}$ in PAW has been measured in previous studies through electron paramagnetic resonance (EPR) spectroscopy [41, 42]. However, this technique requires specialized equipment and techniques. An alternative strategy is to use an indirect reactive probe that responds to the presence of $OH^{\cdot}$ in a manner that is easily observable. Indigo carmine, a common industrial dye (also called acid blue 74), is suitable for this purpose. Its strong absorbance peak around 600 nm is lost upon oxidation of its highly susceptible central C=C bond by $OH^{\cdot}$. The reaction of indigo carmine with $OH^{\cdot}$ to produce isatin sulfonic acid (ISA) is shown in reaction scheme (10) [43, 44].




(10)

Reaction 10 occurs rapidly and irreversibly, with a rate constant $k_{10}$ of $1.8 \times 10^{10}\ M^{-1}\ s^{-1}$ [45], similar to reported rate constants for the reaction between $OH^{\cdot}$ and tyrosine or glutathione [36]. Indigo carmine also reacts on a similarly rapid time scale with $O_3$, $O_2^{\cdot-}$, $OCl^-$, and other strong oxidizers [43, 46]. With this caveat in mind, we provide various observations and calculations that suggest that $OH^{\cdot}$ generated at or near the plasma–liquid interface is the dominant oxidizing reaction partner.

The use of NEAPP to decolorize organic dyes has been investigated in the context of reducing color pollution [47–49] and wastewater abatement [50, 51]. However, previous studies have focused on the rate of decolorization of the dye rather than its implications for measuring oxidizing species. In this work, decomposition of aqueous indigo carmine is used as an indicator of $OH^{\cdot}$ generation and oxidative strength. As noted above, $OH^{\cdot}$ is formed both near the surface and the bulk of the solution. To help isolate one of these sources, we conduct two sets of experiments.

The artificial PAW solution experiments are designed to characterize the degree to which indigo carmine decomposition occurs in the bulk of the solution as a result of the O=NOOH pathway. Measurements of bulk solution pH, $NO_2^-$, and $H_2O_2$ composition following plasma exposure allow the concentrations of these species for the subsequent artificial PAW experiments to be determined. Indigo carmine is then added to a solution containing the same species, but having never been in contact with plasma. By modeling the artificial PAW experiments and confirming that indigo carmine decomposition under these conditions followed known kinetics, we are able to quantify rates of bulk-only decomposition.

In the second set of experiments, rates of indigo carmine decomposition in water solutions exposed to NEAPP are measured. In this set, solutions are either unbuffered or buffered (to pH 7.4 or pH 3.4). Buffering the solution eliminates the time variation of pH, at least in the bulk solution away from the near-surface region. It is possible that the solution buffer is not perfectly effective near the surface in the presence of relatively high fluxes of reactive species. However, these experiments include plasma-induced decomposition in both regions: near the surface and in the bulk solution. By comparing results

from the two sets of experiments, we show the extent to which the indigo carmine decomposition during NEAPP treatment occurs near the plasma–solution interface versus in the bulk solution.

The decomposition products of NEAPP-treated solutions of indigo carmine are analyzed for indications of (a) multiple $OH^{\cdot}$ addition or (b) nitration, and to confirm that the degradation products are consistent with isatin sulfonic acid, as shown in reaction (10). The flux of plasma-generated hydroxyl radicals to the surface is estimated, and compared with the observed levels of indigo carmine degradation.

Overall, this study illustrates the importance of highly reactive species present at the plasma–water interface in determining the effects of NEAPP on adjacent liquid water. These results are consistent with the findings reported in [20, 31], wherein the penetration depth of highly reactive species is modelled for plasma–water systems.

## Methods

### Discharge Apparatus

A simple pin-to-plane direct current (DC) discharge device in air is used for these experiments. The DC discharge device is operated in positive polarity; a positive voltage is applied to the pin electrode (anode), and the grounded electrode is submersed in an aqueous solution, such that the solution surface acts as the cathode. The inter-electrode distance (distance from the pin to solution surface) is fixed at 5 mm. The solution is contained in a glass reservoir with an inner diameter of 29 mm. A copper disc (60 mm diameter, 5-mm thick) is used as the current collector, and the volume of solution treated is 5 mL unless otherwise noted.

A Glassman MK 20P3.7 HV DC power supply is used in voltage control mode to apply a constant voltage to the system. A 10 MΩ ballast resistor is placed in series with the pin electrode; when breakdown of the interelectrode gap occurs, the increased voltage drop across this resistor effectively extinguishes the discharge, resulting in a self-pulsed DC discharge. Importantly, the ballast resistor limits the temperature of the discharge and prevents the transition to thermal spark/arc. Two high voltage probes (Tektronix P6015A) connected to an oscilloscope (Tektronix TDS2024) are used to monitor the voltage at the outlet of the power supply (applied voltage, $V_{app}$) and at the powered electrode ($V_p$), as shown in Fig. 2. It should be noted that the these probes contribute to the overall impedance of the system; these effects are discussed in detail elsewhere [52]. The current to ground ($I$) is calculated from the voltage drop across a 50 Ω resistor, also measured with the oscilloscope. The discharge power is calculated by integrating $I \cdot V_p$ over individual pulses and multiplying by the frequency, using an average of many pulses.

A distinctive feature of positive corona discharges and the discharge used in this work is the presence of a net flow of gas molecules to the surface, a so-called "ionic wind" [29, 53]. This downward gas flow impinging on the treated surface is caused by the acceleration of positive ions towards the cathode, which impart momentum to neutral molecules as well. When the discharge is in contact with a liquid, the impingement of the ionic wind on the liquid both perturbs the surface and induces convection currents in the liquid phase, as shown in a previous model [20]. Visual tests of dye dissolution show that the bulk solution is well-mixed within seconds when concentrated dye is added during

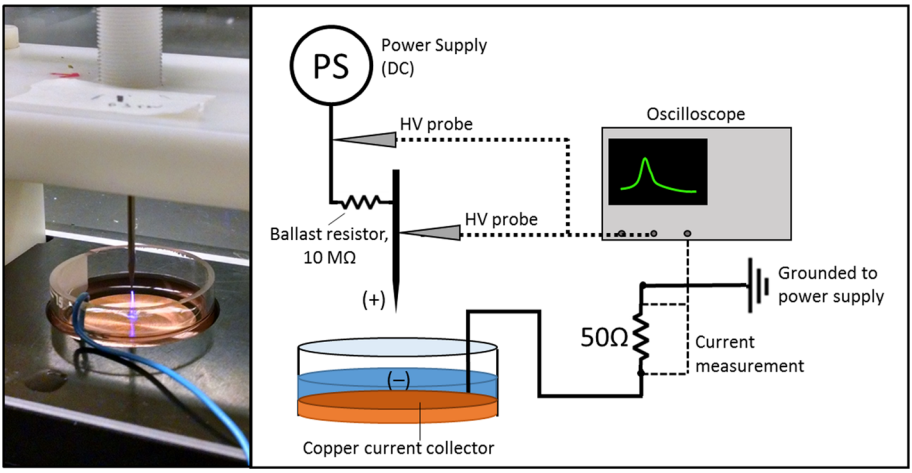



**Fig. 2** Image of the transient spark discharge (*left*) and schematic of the plasma discharge device with equipment for monitoring voltage–current characteristics (*right*)

NEAPP treatment. The solution exposed to the discharge is thus effectively self-mixed up to depths of at least ~15 mm with the current geometry.

One advantage of this type of discharge device is the ability to access multiple discharge modes by varying the electrical parameters and geometry. At a gap distance of 5 mm, a weak streamer discharge is first observed at an applied voltage of around 6 kV. As the voltage is increased to ~8.5 kV, the discharge transitions from a streamer to a "transient spark discharge" (TS) [54]. In this mode, as characterized by Machala et al. [52, 55, 56], a conductive plasma bridges the entire inter-electrode gap with each pulse. Gas temperatures in the transient spark mode are higher than in streamer mode (~550 K for TS vs. ~350 K for streamer); however, the transient spark discharge is still well within the non-thermal regime [57]. The peak current reached is only 60–80 mA, the average current is ~0.3 mA, and a discharge event (lasting ~1 μs) occurs roughly every 67 μs (at a frequency of 15 kHz). This means that during transient spark operation, a discharge occupies the inter-electrode gap for less than 1.5 % of the time. It should be noted that the discharge mode is also dependent on the magnitude of the ballast resistor used. Use of a ballast resistor of 2 MΩ or less results in a constant-current glow discharge in this voltage range.

Typical voltage and current profiles for the transient spark discharge applied in these studies and the low power streamer are shown in Fig. 3. The top traces in Figs. 3a and 3b show the voltage applied by the power supply ($V_{app}$). In this geometry (5 mm inter-electrode distance, 10 MΩ ballast resistor), 9 kV produces a transient spark discharge, while 7 kV results in a weak streamer. The transient spark mode is characterized by the large reduction in the anode voltage during discharge (see second trace in Fig. 3b). The transient spark is also characterized by a step change in discharge frequency, higher peak currents and a longer current pulse duration (~1 μs for transient spark, versus 100–300 ns for streamer). The transient spark discharge is more stable in the presence of the aqueous solution. It is also likely that the more intense discharge results in higher rates of plasma-induced chemistry, and therefore more reactive species generation and greater oxidative power.

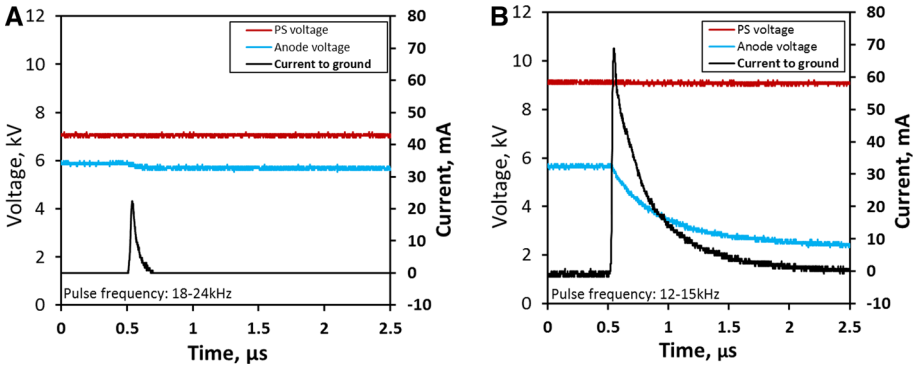



**Fig. 3** Voltage–current characteristics for **a** streamer discharge ($V_{app} = 7$ kV) and **b** transient spark discharge ($V_{app} = 9$ kV). Note the large differences in maximum current and the duration of the current pulse between the two modes

In the present study, the 'base case' NEAPP exposure refers to a 9 kV transient spark discharge unless otherwise noted; the discharge power is 1.4 W, pulse frequencies are 12–15 kHz, and peak current ranges from 50 to 80 mA.

## Chemical Analyses

The solution pH is measured with a Hanna Instruments pH meter (HI 3221, small volume probe). Solution conductivity is measured with a YSI 3200 conductivity instrument. Phosphate ($NaH_2PO_4/Na_2HPO_4$, pH 7.4) and citrate ($H_3C_6H_5O_7/NaH_2C_6H_5O_7$, pH3.5) buffers are prepared to anion concentrations of 4 mM and 10 mM respectively; the initial conductivity of both solutions is 660 µS/cm. A high-conductivity solution of 68 mM NaCl in ultrapure water ($\sigma_{initial} = 7.5$ mS/cm) was used to assess the effects of elevated conductivity while allowing the pH to evolve. All buffers and solutions are made with ultrapure water (18.3 MΩ cm).

The concentration of $H_2O_2$ in solution is determined colorimetrically using the titanium sulfate method [21, 58, 59]. Sodium azide is added to samples immediately following plasma treatment to scavenge nitrites that might react with $H_2O_2$. Note that it was necessary to limit buffer concentrations to 20 mM or less in samples for $H_2O_2$ analysis in order to prevent precipitation of the titanium sulfate reagent, which would interfere with the absorbance measurement.

The concentration of nitrite in solution is measured based on its reaction with Griess reagent (Fluka Analytical). Samples and standard solutions are developed with reagent for 30 min, and nitrite concentrations are calculated from the absorbance at 548 nm over a 1 cm path length, based on a calibration curve. Samples for nitrite analysis are diluted to a concentration of 200 µM or less in order to remain within the linear absorbance regime. Measured concentrations of nitrite are not significantly affected by the presence of $H_2O_2$ during the development period; at such dilute concentrations, the rate of $NO_2^-$ reaction with $H_2O_2$ is negligible compared to reaction with the colorimetric reagents, which are present in large excess.

Indigo carmine (Spectrum Chemicals) is stored in the dark, and is checked regularly for degradation. The concentration of indigo carmine in buffered solutions prior to plasma exposure is 100 µM. Indigo carmine concentrations are determined by absorbance at

600 nm and comparison to a calibration curve. The extinction coefficient of indigo carmine is found to be 19,800 $M^{-1}$, consistent with values reported in the literature [43, 60, 61]. All absorbance measurements are made using an Agilent Cary-60 UV/vis spectrometer with a 1 cm path length.

Mass spectrometry (ESI–MS, Agilent QTOF) is used to identify the decomposition products of indigo carmine. Samples are directly injected into the instrument in both negative and positive ion mode, based on the presence of sulfate and amine groups in the parent molecule. In negative ion mode, unreacted indigo carmine is a doubly charged ion and appears at an m/z of 210.

$^1$H NMR is also used to identify the degradation products of indigo carmine exposed to the transient spark discharge. Samples are run on a Bruker 500 MHz magnet. A solvent-suppression sequence (ZGPR) is used to saturate the peak corresponding to water. $D_2O$ (10 %) is added to enable locking, and 1 mM (trimethyl-silyl)propionic acid ($\delta = 0$ ppm) is used as an internal standard.

## Results and Discussion

An appropriate set of operating conditions for this plasma source is determined by investigating the effects of applied voltage, solution conductivity, and pH on discharge power and $H_2O_2$ generation. Four solutions are treated with NEAPP at applied voltages ranging from the onset of a visible discharge (roughly 7 kV) to 11 kV for a 5-min period. The solution conductivity is varied over 3 orders of magnitude (NaCl is added to increase solution conductivity); solution pH is allowed to evolve with NEAPP exposure, or is fixed at 3.5 or 7.4 through the use of buffers. Discharge power and $H_2O_2$ production in the solution are expected to be related to the relative $OH^{\cdot}$ density near the plasma–liquid interface [39]; thus, these parameters are evaluated as a rough indicator of oxidizing strength.

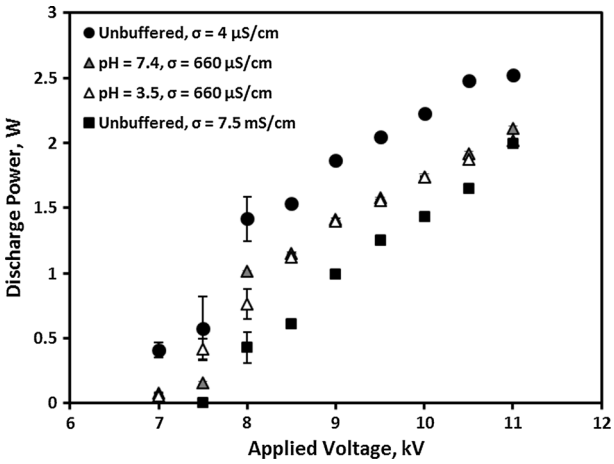



**Fig. 4** Dependence of discharge power on voltage showing the effects of solution pH and conductivity. Discharge power was determined from the period-averaged product of current (to ground) and anode voltage. Discharge power increases with decreasing solution conductivity at constant applied voltage in the transient spark mode but is independent of pH

The discharge power as a function of applied voltage ($V_{app}$) is shown in Fig. 4. The discharge power generally increases proportionally with applied voltage, as expected, but is also a function of solution conductivity at a given voltage. A discontinuity is observed around 8–8.5 kV, corresponding to the mode transition from the streamer to transient spark discharge mode. In the transient spark regime (voltages greater than about 8 kV), there is a counterintuitive dependence of dissipated power on solution conductivity: power increases at a given voltage for lower solution conductivity. This is discussed in greater detail below.

Figure 5 shows that $H_2O_2$ production in solution is primarily a function of applied power and is relatively independent of either solution conductivity or pH. In all test solutions, the $H_2O_2$ concentration varies linearly with power after the transition to transient spark mode at discharge powers above about 1 W. Within the transient spark mode, the rate of $H_2O_2$ generation may be slightly reduced at the highest solution conductivity tested (NaCl added, 7.5 mS/cm) relative to the other two solutions. This may be related to the reduction in solution resistance, which may lead to reduced "spreading" of the discharge over the liquid surface [62, 63], or to the generation of chlorinated species ($OCl^-$) in solution and subsequent reaction with $H_2O_2$ at increased discharge power.

The observed increase in $H_2O_2$ generation with increased power over short time scales is likely due to a corresponding increase in various plasma-induced reactions leading to the formation of $OH^{\cdot}$ in the gas phase (i.e. reactions (1–4) [30]. Laser-induced fluorescence (LIF) studies of a positive pulsed DC discharges have shown that $OH^{\cdot}$ density is roughly proportional to power dissipated by the discharge in similar systems [64]. Because $H_2O_2$ is predominantly formed from recombination of $OH^{\cdot}$ in the gas phase through reaction (1) [37], increased $OH^{\cdot}$ density near the plasma–liquid interface due to increased discharge power likely results in increased production of $H_2O_2$.

The pH of the target solution has no discernible effect on the discharge power (at a given applied voltage) or $H_2O_2$ production over the 5-min period examined. The effect of solution conductivity is less straightforward. Figure 4 shows that discharge power declines as solution conductivity increases over the range of conductivities considered.

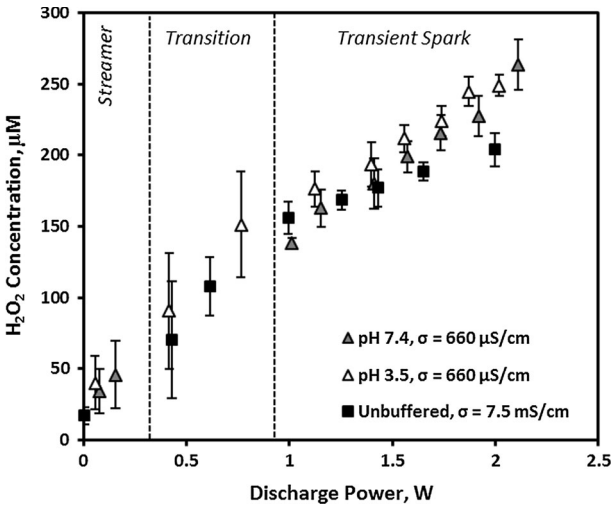



**Fig. 5** Generation of $H_2O_2$ as a function of discharge power after 5 min of discharge operation. $H_2O_2$ increases linearly with increasing discharge power

Understanding this effect will require further study of the interactions between the gas phase plasma and the liquid electrolyte solution. We note that solution conductivity appears to have a minor effect on total $H_2O_2$ production over the range of conductivities considered, particularly in the transient spark mode. Nevertheless, in the experiments described below, we maintain an effectively constant solution conductivity through the use of buffers to minimize the effects of this parameter on plasma–solution interactions.

The evolution of reactive species in NEAPP-treated water as a function of time is investigated. The initial experiments on the chemical changes occurring in solution are performed with ultrapure water without buffers. The solution equilibrates in air prior to NEAPP treatment; resulting in an initial conductivity of 4 μS/cm and a pH of ~5.7 due to dissolved atmospheric $CO_2$. In these and subsequent experiments, a voltage of 9 kV is applied to the system, resulting in a transient spark discharge and 1.4 W dissipated by the plasma.

As ultrapure water is treated with NEAPP, the solution acidifies through the absorption of plasma-generated $NO_3^-$ and $NO_2^-$ into the liquid, and through the subsequent production of $NO_3^-$ from $NO_2^-$ in the liquid phase. The measured changes in pH and in the concentrations of $H_2O_2$ and $NO_2^-$ during NEAPP treatment of initially ultrapure water are shown in Fig. 6. Under acidic conditions, nitrite is converted to nitrate through the disproportionation/degradation of $HNO_2$ according to reactions (11)–(13) [21, 27].

$$NO_2^- + H^+ \leftrightarrow HNO_2 \tag{11}$$

$$2HNO_2 \rightarrow NO^{\cdot} + NO_2^{\cdot} + H_2O \tag{12}$$

$$2NO_2^{\cdot} + H_2O \rightarrow NO_3^- + NO_2^- + 2H^+ \tag{13}$$

Note that reaction (11) describes the equilibrium of $HNO_2/NO_2^-$ in solution ($pK_a = 3.39$). Below the $pK_a$ of nitrous acid, reactions (11)–(13) will lead to essentially complete conversion of $HNO_2$ and $NO_2^-$ into $NO_3^-$ and $H^+$. If $H_2O_2$ is present as well, reactions (7)–(9) will lead to $OH^{\cdot}$ formation. This explains the trends observed in Fig. 6; the concentrations of $H_2O_2$ and $NO_2^-$ initially increase, but level off or decrease as the solution is further acidified and rates of pH-dependent reactions (7–9, 11–13) increase and consume them.

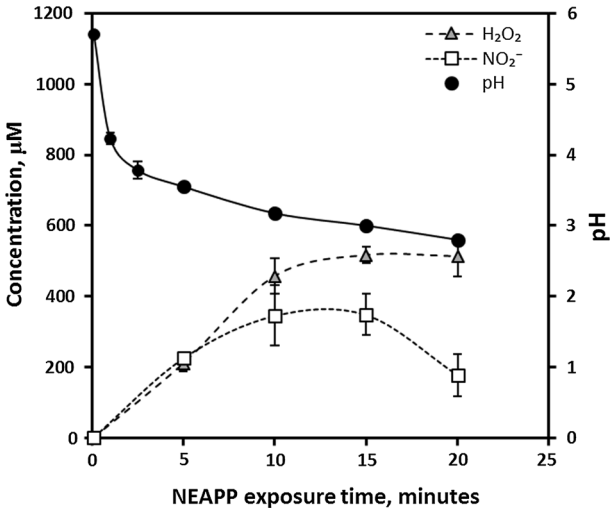



**Fig. 6** Evolution of pH, $[H_2O_2]$ and $[NO_2^-]$ in unbuffered, initially ultrapure water with NEAPP exposure

Furthermore, the absorption of $NO_2^-$ and $NO_3^-$ increases the solution conductivity of ultrapure water significantly. The addition of 100 μM of $NO_3^-$ increases the solution conductivity by ~12 μS/cm; for ultrapure water solutions, absorption of $NO_3^-$ can change the solution conductivity by order of magnitude above the initial level (4 μS/cm). As noted above (see Fig. 4), changes in solution conductivity can affect discharge V–I characteristics. In the results presented below, pH is fixed by buffers and solution conductivity is maintained near 660 μS/cm.

The generation of $H_2O_2$ and $NO_2^-$ in acidic (pH 3.5) and neutral (pH 7.4) buffered solutions is shown in Fig. 7. Figure 7 illustrates that under neutral conditions (pH 7.4), the concentrations of $H_2O_2$ and $NO_2^-$ in solution grow at a steady rate. The linear increase in the concentrations of these species (consistent with the trends observed by Lukes et al. [21]) suggests that they are generated by the discharge at a relatively constant rate. However, accumulation of both $H_2O_2$ and $NO_2^-$ in this system begins to slow after 5 min of NEAPP exposure under acidic conditions. As the concentrations of $H_2O_2$ and $NO_2^-$ in the bulk liquid accumulate to levels above 500 μM at pH 3.5, the rate at which these species are consumed through reaction (7) becomes significant. In previous work by Lukes et al., concentrations of $H_2O_2$ and nitrite did not exceed 500 μM; thus, in those studies, rates of species generation remained constant over the entire range of NEAPP exposures studied, even under acidic conditions [21]. Under neutral conditions (pH 7.4), the concentrations of $H_2O_2$ and $NO_2^-$ continue to increase linearly even at millimolar concentrations.

With the concentrations of $H_2O_2$ and $NO_2^-$ generated by this device known, the rate of $OH^{\cdot}$ production in plasma-activated water solutions (PAW) through the O=NOOH pathway can be modeled. The percentage of O=NOOH decomposing to $OH^{\cdot}$ through reaction (8) and contributing to indigo carmine oxidation is evaluated experimentally by combining $H_2O_2$, $NO_2^-$ and nitric acid at the concentrations present after 20 min of NEAPP treatment. We refer to these solutions as 'artificial PAW.' The rate of indigo carmine oxidation in these solutions is then monitored spectrophotometrically. Because no plasma is involved—and therefore no species beyond $H_2O_2$, $NaNO_2$ and $HNO_3$—$OH^{\cdot}$ produced through O=NOOH degradation is the only oxidizing species present. These artificial PAW experiments are modeled to confirm the kinetics of this system and determine the relation between O=NOOH production and indigo loss. The artificial PAW and model results are

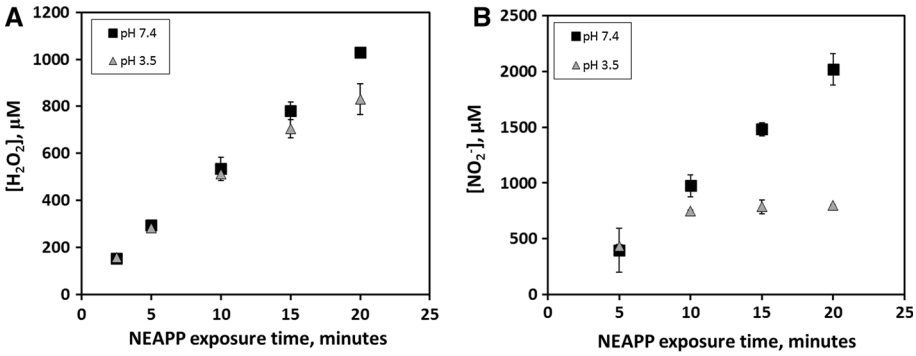



**Fig. 7** Concentration of **a** hydrogen peroxide and **b** nitrite as a function of plasma exposure time under conditions of controlled pH and solution conductivity (660 μS/cm)

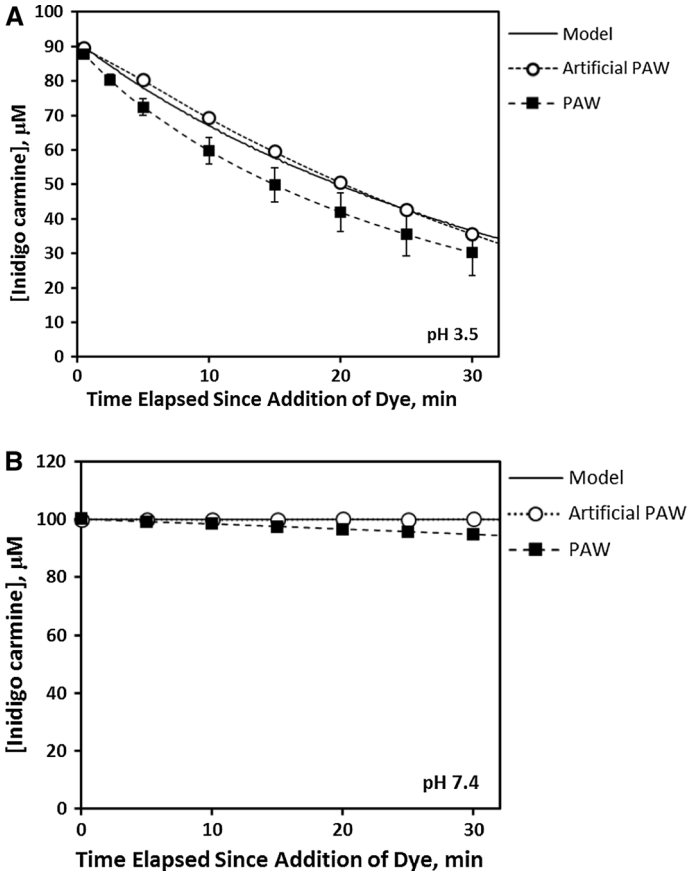



**Fig. 8** Degradation of indigo carmine in plasma-activated water, PAW (*filled square*), artificial PAW (*open circle*), or predicted by the model (*solid line*) described in Online Resource 1 at **a** pH3.5 and **b** pH 7.4. Solutions shown in **a** initially contained 800 μM $H_2O_2$ and 800 μM $NO_2^-$; solutions in **b** contained 1 mM $H_2O_2$ and 2 mM $NO_2^-$, as measured in samples treated with NEAPP for 20 min

shown for comparison in Fig. 8. In brief, this model of the decline in indigo carmine concentration assumes that:

a. The solution is well mixed.
b. $H_2O_2$ and $NO_2^-$ react to produce O=NOOH per reaction (7).
c. O=NOOH decomposes through a pathway described by reactions 8–9 with some fraction, *b*, of the O=NOOH producing $OH^{\cdot}$ through reaction 8. This parameter (*b*) is determined by fitting the model to the experimental artificial PAW data.
d. Each $OH^{\cdot}$ that is produced reacts with a single molecule of indigo carmine.
e. All indigo carmine degradation occurs due to reaction with $OH^{\cdot}$, per the kinetics shown in reaction 10.

The artificial PAW solutions were vortexed immediately following $H_2O_2$ addition to ensure the solutions were well mixed. At a controlled pH of 3.5, the degradation of indigo carmine in artificial PAW was in good agreement with the kinetic rate data reported in [21] with a rate constant for reaction (7), $k_7$, equal to $1.1 \times 10^3$ $M^{-2}$ $s^{-1}$.

A branching coefficient $b$ of 0.24 [see reactions (7–9)] yielded the best fit to the artificial PAW data as shown in Fig. 8, and is in good agreement with the range of branching coefficients reported in the literature for O=NOOH decomposition ($\sim$ 30 %) [40]. The system of equations solved in this model is described in greater detail in Online Resource 1. In solutions buffered to neutral pH (pH 7.4), artificial PAW with identical concentrations of $H_2O_2$, $NO_2^-$ and nitric acid in solution result in no significant oxidation of indigo carmine, also consistent with the model of $OH^{\cdot}$ formation through the O=NOOH pathway. This confirmed that the O=NOOH pathway to form $OH^{\cdot}$ is not active at a pH of 7.4. These artificial PAW experiments provide the data needed to assess the importance of bulk solution decomposition of indigo carmine occurring in NEAPP-treated solutions.

In order to examine the rate of indigo carmine decomposition occurring only due to species in the bulk—and not at the surface—of the plasma activated water solution (PAW), indigo carmine is added to the solution immediately after NEAPP treatment. The concentrations of $H_2O_2$ and $NO_2^-$ following NEAPP treatment (and just prior to indigo carmine addition) are the same as the concentrations used in the artificial PAW experiments. The decay curve of indigo carmine added to pH 3.5-buffered PAW solutions (treated for 20 min) is similar to the artificial PAW experiments, as shown in Fig. 8. These experiments show that the indigo carmine decomposition reactions measured in the artificial PAW experiments are very similar to those that occur in the bulk of the PAW solutions.

This set of results also implies that $OH^{\cdot}$ produced through O=NOOH decomposition is in fact the primary oxidizing agent in these plasma-activated water solutions in the absence of active plasma. If significant concentrations of dissolved $O_3$ were present at the end of NEAPP treatment, for example, the $O_3$ would react with indigo carmine almost instantly ($k > 10^7$ $M^{-1}$ $s^{-1}$, [43]), rather than over the course of several minutes as shown in Fig. 8. Although some loss of indigo carmine was observed when indigo carmine was immediately added to pH 7.4 PAW solutions, the limited extent of indigo carmine oxidation (<6 μM over 30 min) indicates that PAW solutions do not contain significant concentrations of other long-lived, non-acid catalyzed oxidizing species.

The observed degradation of indigo carmine solutions treated directly with NEAPP at pH 3.5 and pH 7.4 as a function of exposure time is shown in Fig. 9. These data are

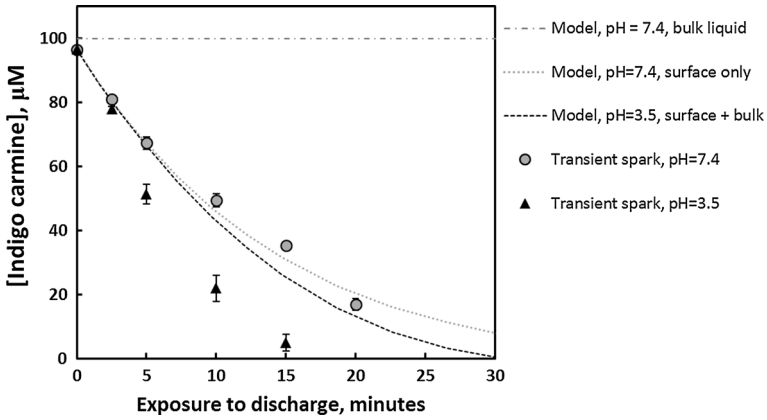



**Fig. 9** Degradation of indigo carmine by transient spark discharge, experimental versus predicted from reactions of RONS in bulk liquid under neutral (pH 7.4) and acidic (pH 3.5) conditions

compared to the loss of indigo carmine predicted to occur due to reaction with $OH^{\cdot}$ in the bulk solution (i.e. produced through the O=NOOH pathway, reactions 7–9) and/or due to reactions at or near the surface.

The rate of indigo carmine oxidation that is attributed to reaction with $OH^{\cdot}$ or other oxidizing species at the plasma–liquid interface ("surface only") is determined by estimating the effective rate of indigo carmine oxidation from the initial loss during NEAPP treatment at pH 7.4. At a pH of 7.4, the O=NOOH pathway and any other reactions arising from stable RONS in the bulk solution (specifically $H_2O_2$, $NO_2^-$, and $NO_3^-$) do not significantly oxidize indigo carmine, as was shown in Fig. 8. Thus, the degradation of indigo carmine by NEAPP at neutral pH is attributed to reaction with $OH^{\cdot}$ or other highly reactive species at or near the plasma–liquid interface. As previously discussed, these species are not expected to diffuse beyond several microns; these effects are therefore described as surface reactions. The first time point at 2.5 min is used as an initial estimate since at this point the concentration of indigo carmine in the bulk solution is not significantly depleted. The initial rate of indigo loss is thus estimated to be approximately 6.2 μM/min ($1.0 \times 10^{-7}$ M/s).

$$d[\mathrm{IC}]_{\mathrm{initial}}/dt = -6.2\,\mu\mathrm{M/min} = 1.0 \times 10^{-7}\,\mathrm{M/s} \approx -k_{10}[\mathrm{OH}^{\cdot}]_{\mathrm{surf}}[\mathrm{IC}]_{\mathrm{initial}} \tag{14}$$

Assuming that the concentration of $OH^{\cdot}$ near the surface is roughly constant, the concentration of $OH^{\cdot}$ can be lumped into the rate constant to estimate an effective rate constant, $k_{eff}$, with units of $s^{-1}$. Knowing the initial concentration of indigo carmine, 100 μM ($10^{-4}$ M), we solve Eq. 15 to estimate $k_{eff}$.

$$\begin{aligned} d[\mathrm{IC}]_{\mathrm{initial}}/dt &= -6.2\,\mu\mathrm{M/min}\left(1.0 \times 10^{-7}\,\mathrm{M/s}\right) \approx -k_{10}[\mathrm{OH}^{\cdot}]_{\mathrm{surf}}[\mathrm{IC}]_{\mathrm{initial}} \\ &\approx -k_{\mathrm{eff}}[\mathrm{IC}]_{\mathrm{initial}} \rightarrow k_{\mathrm{eff}} \approx 1.0 \times 10^{-3}\,\mathrm{s}^{-1} \end{aligned} \tag{15}$$

Because this rate is calculated from a difference and not measured instantaneously, it does not take into account the dependence on indigo carmine concentration. Fitting this expression to the initial data, we find $k_{eff} = 1.2 \times 10^{-3}\ s^{-1}$. The plasma-induced degradation of indigo carmine due to surface reactions only (pH 7.4) is thus written as:

$$d[\mathrm{IC}]/dt = -k_{\mathrm{eff}}[\mathrm{IC}] \quad \text{where } k_{\mathrm{eff}} = 1.2 \times 10^{-3}\,\mathrm{s}^{-1} \tag{16}$$

It is apparent that loss of indigo carmine at pH 7.4 greatly exceeds that expected from reactions in the bulk solution (namely the O=NOOH pathway). At a pH of 7.4, the O=NOOH pathway (reactions 7–9) should be inactive, although the local pH very close to the plasma–liquid interface may differ significantly from the bulk.[1] Clearly, an additional source of $OH^{\cdot}$ (beyond the O=NOOH pathway in the bulk solution) is required to explain the oxidation observed during NEAPP treatment. The following effects may be responsible: (a) $OH^{\cdot}$ or other highly oxidizing species generated directly by the plasma react with indigo carmine at or very near the plasma–liquid interface; (b) the interaction of NEAPP and NEAPP-generated species with the liquid creates a thin surface region with elevated concentrations of $H_2O_2$ and $NO_2^-$, and/or where the buffer is overcome and pH is

[1] In simple experiments in which methyl red solutions (buffered to pH 7.4) were treated with NEAPP, no color change (indicating a gradient in pH) was observed near the surface or anywhere in solution. Solutions were treated with NEAPP until the buffer was overcome, at which point the color uniformly changed from yellow to red. This is in contrast to experiments performed by Witzke et al. [79] (in the absence of convection), in which spatial gradients in pH were observed at the plasma-liquid interface. This suggests that any gradients in pH that exist are restricted to very near the surface.

significantly reduced, leading to an accelerated rate of $OH^{\cdot}$ formation through the O=NOOH pathway; (c) indigo carmine at/near the surface is degraded by UV or other energetic plasma species. Note that in two of these cases, $OH^{\cdot}$ is the primary oxidizing species responsible for modifying the target (indigo carmine).

When pH 3.5, indigo carmine is degraded by both direct interaction with plasma and through the O=NOOH pathway in the bulk solution. Unlike the rate of indigo decomposition at the surface, the rate of degradation in the bulk liquid is dependent on the bulk concentrations of $H_2O_2$ and $NO_2^-$ and will therefore change with time. The rates of generation of $H_2O_2$ and $NO_2^-$ during plasma treatment were estimated from Fig. 7. The rates for the generation of $H_2O_2$ and $NO_2^-$ are determined from the measured concentrations of these species at up to 5 min of plasma treatment, and represent the net increase in bulk liquid concentration of these species in the absence of consumption reactions. At later time points, the rate of generation is convoluted by significant loss of these species through the O=NOOH pathway (Eq. 7).

$$r_H_2O_2 = 1.0 \times 10^{-6}\,\mathrm{M/s} \tag{17}$$

$$r_NO_2^- = 1.6 \times 10^{-6}\,\mathrm{M/s} \tag{18}$$

The evolution of $H_2O_2$, $NO_2^-$, and O=NOOH is thus modeled from the rates of generation and consumption using the following the system of equations:

$$d[H_2O_2]/dt = r_H_2O_2 - k_7[H_2O_2]\left[NO_2^-\right][H^+] \tag{19}$$

$$d\left[NO_2^-\right]/dt = r_NO_2^- - k_7\left[H_2O_2\right]\left[NO_2^-\right]\left[H^+\right] \tag{20}$$

$$d[O = NOOH]/dt = k_7[H_2O_2]\left[NO_2^-\right][H^+] - 0.13 + 0.87[H^+] \cdot [O = NOOH] \tag{21}$$

Similar to the artificial PAW analysis, the rate of indigo carmine loss through the O=NOOH pathway occurring in the bulk solution is related to the rate of O=NOOH formation.

$$(d[IC]/dt)_{bulk,initial} = -k_{10}[IC]_{initial}[OH^{\cdot}]_{bulk} = b \cdot (0.13 + 0.87[H^+])[O = NOOH] \tag{22}$$

Because the initial concentration of indigo carmine is known, this expression can be rearranged to write an expression for $-k_{10}[OH^{\cdot}]_{bulk}$, which is dependent on the concentration of O=NOOH:

$$-k_{10}[OH^{\cdot}]_{bulk} = b \cdot (0.13 + 0.87)[H^+] \cdot [O = NOOH]/[IC]_{initial} \tag{23}$$

$$(d[IC]/dt)_{bulk} = -k_{10}[IC][OH^{\cdot}] = b \cdot (0.13 + 0.87[H^+]) \cdot [O = NOOH] \cdot [IC]/[IC]_{initial} \tag{24}$$

Thus, the total loss of indigo carmine at pH 3.5 from both surface and bulk liquid oxidation of indigo carmine is modeled as:

$$d[IC]/dt = -k_{eff}[IC] - b \cdot (0.13 + 0.87[H^+]) \cdot [O = NOOH] \cdot [IC]/[IC]_{initial} \tag{25}$$

The model of both surface-mediated indigo carmine loss (with $k_{eff} = 1.2 \times 10^{-3}\ s^{-1}$) and loss due to the O=NOOH pathway (reactions in bulk solution) at pH 3.5 is compared to the experimental transient spark data in Fig. 9. The loss of indigo carmine at pH 7.4 (model and experimental data) are shown in grey for comparison. The model is described in

additional detail in Online Resource 1. It is evident that the loss of indigo carmine exposed to plasma at a pH of 3.5 is significantly greater than that predicted by the model that includes both surface and bulk solution effects. This suggests an increase in the rate of reaction of indigo carmine at the surface when the solution is buffered to pH 3.5 (with a buffer concentration of 10 mM citrate/citric acid). The solution pH does not affect the properties of the transient spark discharge, as shown in Figs. 4 and 5. This effect must therefore arise from conditions in the liquid phase, such as increased concentrations of certain RONS and/or decreased pH (the buffer is locally overcome) very near the plasma–liquid interface, such that the rate of indigo carmine decomposition due to acid-catalyzed reactions in this region is increased beyond that observed when the pH near the surface is held at 7.4. Both series of data suggests that interactions at the plasma–liquid interface (rather than reactions in the bulk liquid) play a dominant role in plasma–liquid chemistry.

The distance that $OH^{\cdot}$ and other highly reactive species penetrate into an adjacent phase (and depth of the surface reaction zone) is difficult to determine precisely due to their fast reaction kinetics and hence strong dependence on the concentrations of a suite of possible reaction partners. However, most published models of plasma–liquid interactions show that the concentration of $OH^{\cdot}$ entering from the gas phase drops by orders of magnitude over a few microns or tens of microns from the surface. For example, calculations by Lindsay et al. [20] suggest a 9-order of magnitude decrease in $OH^{\cdot}$ concentration within 50 μm of the surface in unbuffered water. A model of DBD-treated tissue by Tian and Kushner [31] indicates that the liquid-phase $OH^{\cdot}$ concentration becomes negligible after a similar distance. Biological reviews report $OH^{\cdot}$ diffusion distances that are even smaller: on the order of "a small protein" [65], perhaps 10–100 nm. Experimental studies of X-ray irradiation of mammalian cells suggest an average intracellular diffusion distance for $OH^{\cdot}$ of 10 nm [66]. Thus, the near-surface region in which indigo carmine and $OH^{\cdot}$ react is expected to extend between $\sim$100 nm to possibly several tens of micrometers into the liquid phase. In Online Resource 2, we illustrate that the observed rates of indigo carmine oxidation are not inconsistent with arrival of $OH^{\cdot}$ from the gas phase (assuming concentrations produced by a positive DC discharge in humid air, and given reasonable assumptions about the rates of mass transfer of $OH^{\cdot}$ from the gas phase to the liquid).

It should be noted that the previous point does not address the possibility that transport of the dye to the surface may in fact be limiting. Just as the concentrations of $OH^{\cdot}$ and other plasma-generated RONS may differ significantly from their bulk values near the plasma–liquid interface, indigo carmine and the buffers used to control pH may be depleted in this region. It is almost certainly the case that thin boundary layers with relatively high concentrations of oxidizing species at the surface create areas of depletion of such species. In this case, diffusive or convective transport of the dye to this region from the bulk may be limiting, although this is expected to occur on time scales of milliseconds or less. Mass transport effects may therefore play an important role in determining the extent of plasma modification of species in the liquid phase.

In order to identify the decomposition products of indigo carmine and assess the extent to which $OH^{\cdot}$ subsequently reacts with these species, plasma-treated indigo carmine solutions are analyzed with mass spectrometry and $^{1}H$ NMR spectroscopy. Electrospray ionization mass spectrometry of NEAPP-treated solutions containing indigo carmine yield a dominant signal at m/z = 226 in negative ion mode, consistent with isatin sulfonic acid (ISA) as the main decomposition product, as shown in reaction (10). Additional small peaks were observed at m/z = 244 (ISA + O) and 147 (ISA-$SO_3$ + H) in negative ion mode, and at m/z = 143 and 190 (unassigned) in positive ion mode. No evidence of nitrated species was observed.

NEAPP-treated indigo carmine solutions are also analyzed with $^1$H NMR to confirm our assignment of the peak observed at m/z = 226 to ISA. Indigo carmine standards produce $^1$H NMR peaks at $\delta = 8.04$(s), 7.74(d) and 6.86(d) ppm. NEAPP-treated indigo carmine solutions showed $^1$H signals consistent with indigo carmine, and also at $\delta = 7.14$(t) and 7.94(m). The latter correspond to the $^1$H NMR spectra from ISA standards, providing further evidence that ISA is the primary decomposition product of indigo carmine under these conditions. No peaks corresponding to other possible decomposition products, such as sulfoanthranilic acid, malic acid, or tartaric acid [51, 67] are observed. Representative $^1$H NMR spectra are shown in Fig. 10. Significantly more indigo carmine is degraded at pH 3.5 relative to pH 7.4, as is seen from comparing Fig. 10a (NEAPP-treated pH 3.5 solution) to Fig. 10b (NEAPP-treated pH 7.4 solution). Due to the relatively high initial concentration of indigo carmine used in the $^1$H NMR studies (2 mM vs. 100 μM), all samples contained high concentrations of unreacted indigo carmine, even after 30 min of NEAPP treatment.

If all indigo carmine lost is oxidized by NEAPP to ISA with no subsequent reaction, loss of 1 mM indigo carmine should be accompanied by a 2 mM increase in the ISA concentration. Integration of the peaks assigned to indigo carmine (6.86 ppm) and ISA (7.14 ppm) in Fig. 10 indicates that ISA concentrations are roughly 60 % of the expected levels. This suggests that further degradation of indigo carmine (perhaps all the way to $CO_2$, given the lack of other products observed) does in fact occur. Thus, competition for $OH^{\cdot}$ or other oxidizing species from ISA and other degradation products should be considered in analyzing such systems.

Despite the above caveat, both analyses show clear evidence of isatin sulfonate as the main decomposition product and no other products were observed in significant concentrations. The production of ISA is consistent with $OH^{\cdot}$ as the primary near-surface oxidizing agent, but is not definitive proof that this is the sole active agent.

Other possible oxidizing species include ozone ($O_3$), the superoxide anion ($O_2^-$), and singlet oxygen ($^1O_2$). Ozone is known to be produced in the gas phase by positive polarity DC discharges, but its reaction with indigo carmine is too rapid to account for the observed bulk decomposition when indigo is added to PAW after plasma treatment. If $O_3$ were

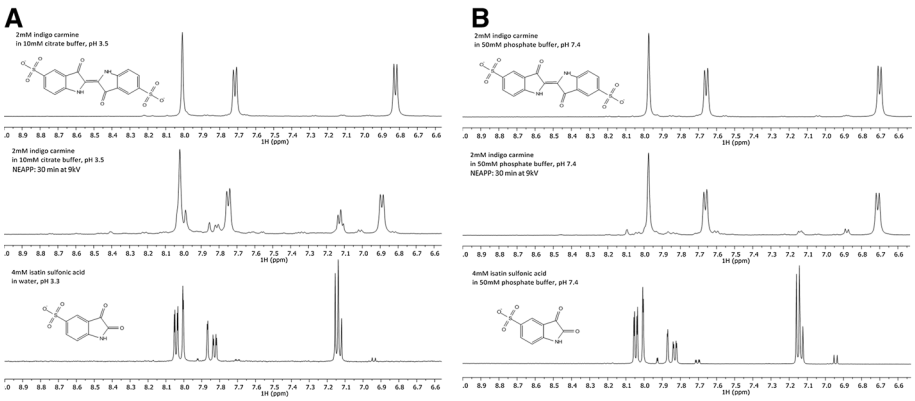



**Fig. 10** $^1$H NMR spectra of transient spark NEAPP-treatment (9 kV, 30 min) of indigo carmine solutions in **a** 10 mM citrate buffer, pH 3.5 and **b** 50 mM phosphate buffer, pH 7.4. The conversion of indigo carmine to isatin sulfonic acid (ISA) is evidenced by the presence of signal at the chemical shifts corresponding to the ISA control

present in solution following NEAPP exposure, it would react with the indigo carmine immediately upon addition of the dye, rather than over the course of several minutes. Furthermore, the elevated gas temperatures of the transient spark discharge ($\sim$550 K) are expected to produce less ozone than low-power discharge modes, as more $NO_x$ species are produced [7] and ozone is decomposed at a higher rate [68].

The superoxide radical has been shown to be generated by plasma jets and negative polarity dc discharges [37, 69, 70]. However, $O_2^{\cdot-}$ formation in the gas-phase may be less likely in positive polarity dc discharges such as the one used in this study, in which water–vapor cluster positive ions (i.e. $H_3O^+[H_2O]_n$) are the dominant charge carriers [71, 72]. Furthermore, the formation of significant quantities of $O_2^{\cdot-}$ at the plasma–liquid surface is unlikely given that the water acts as the cathode. Nevertheless, we cannot exclude the possibility that $O_2^{\cdot-}$ is playing a significant role near the interface without additional studies. At pH values significantly above the pKa of O=NOOH (6.8) the production of superoxide radicals from the decomposition of O=NOOH is expected to play a more significant role.

Singlet oxygen ($^1O_2$) is also a possible oxidant that has recently received a great deal of attention in the plasma community due to recent evidence that it can be an important active species in cell treatment studies [69, 73, 74]. However, previous studies suggest that indigo carmine does not react with singlet oxygen to a significant extent [75, 76].[2] Considering that the rate of indigo carmine oxidation by singlet oxygen is likely orders of magnitude below the reported rate of physical quenching ($3.2 \times 10^8$ $M^{-1}$ $s^{-1}$) and comparing this to the rate of indigo carmine oxidation by hydroxyl radical, $1.8 \times 10^{10}$ $M^{-1}$ $s^{-1}$, significantly more $^1O_2$ must be present (perhaps 3–4 orders of magnitude or more) near the plasma–liquid surface to produce the same rate of indigo carmine loss than if the oxidizing species were $OH^{\cdot}$.

In this system, the evaporation and ionization of water at the plasma–liquid interface is expected to produce significant levels of $OH^{\cdot}$, based on the concentrations produced by a similar device as determined by LIF [64, 77, 78] and the observed concentrations of $H_2O_2$ generated in solution. Given our present knowledge of the system, $OH^{\cdot}$ is likely to be the dominant oxidizer of indigo carmine. However we reiterate that this has not been proven. In this work, we investigate the origin of the observed oxidation of a model compound (indigo carmine), and highlight the importance of the plasma–liquid interface in specie modification. The next logical step is to assess the identity of ROS present ($OH^{\cdot}$ vs. $^1O_2$ or $O_2^{\cdot-}$) through direct measurements with EPR and with competitive kinetic experiments. A more detailed understanding of the near-surface plasma–water chemistry must await future work in which known scavengers or selected reaction partners can be added to react selectively with various possible oxidizing agents [74].

[2] The rate at which $^1O_2$ is quenched (reduced to the triplet state) by indigo carmine is $3.2 \times 10^8$ $M^{-1}$ $s^{-1}$ per the results of Gandra et al. [76]. Moreover, the quenching of $^1O_2$ observed in these experiments was stated to occur through "physical quenching", or processes that convert $^1O_2$ to $^3O_2$ by reducing the energy state of indigo carmine rather than changing its structure ("chemical quenching"). Oxidation products of indigo carmine were only observed after extended irradiation periods (several hours of $^1O_2$ production).

## Conclusions

We demonstrate that a self-pulsed point-to-plane DC discharge, and specifically the transient spark discharge in air, offers a simple, accessible platform for studying plasma/liquid interactions. Initially, the accumulation of stable plasma-generated species ($H_2O_2$ and $NO_2^-$) in the liquid phase proceeds linearly with NEAPP exposure at constant pH in the absence of competing reactions. Under acidic conditions, subsequent reactions in the liquid phase begin to consume significant quantities of $NO_2^-$ and $H_2O_2$ once these species accumulate to concentrations greater than about 500 μM.

Studies of the effects of $H_2O_2$ and $NO_2^-$ on indigo carmine are consistent with the established kinetics of $OH^{\cdot}$ production through the O=NOOH pathway, with 24 % of O=NOOH going to $OH^{\cdot}$ generation and the oxidation of indigo carmine. When indigo carmine is added to plasma-activated solutions, the observed decay rate follows that observed when $H_2O_2$ and $NO_2^-$ are added to solution without plasma exposure. This supports the conclusion that NEAPP treatment does not generate significant concentrations of other, long-lived reactive species in the buffered solutions investigated.

Experiments using indigo carmine as an indicator of oxidative strength during NEAPP exposure demonstrate that the activity of $OH^{\cdot}$ (or other highly oxidizing species) near the plasma–liquid interface dominates the effects of NEAPP treatment. This result is striking given the small area of contact between the discharge and liquid (<1 $mm^2$), and highlights the importance of gas–liquid transport phenomena in these systems. The observed effects could be due to either (a) direct reaction with plasma-generated species, most likely $OH^{\cdot}$, at the interface, (b) locally elevated concentrations of $H_2O_2$, $NO_2^-$, and/or local reduction in the pH at the surface, leading to acceleration of the O=NOOH pathway in this region; or (c) the effect of plasma-generated species such as UV and/or ions directly on indigo carmine. Although these hypotheses await confirmation, our initial results are consistent with $OH^{\cdot}$ being the primary oxidizing agent.

#### Compliance with Ethical Standards

**Conflict of interest** The authors declare that they have no conflict of interest.

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